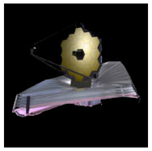

3207 - Lifting the Veil: An Open Source Haze Model for Exoplanet Atmospheric Characterization

Cycle: 2, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

Clouds and hazes are common in the atmospheres of exoplanets. These small particles impact atmospheric radiative transfer and chemistry, and impede our efforts to characterize atmospheres through remote sensing. The extensive, continuous wavelength coverage and greater precision of JWST will likely reveal further impacts of clouds and hazes, including their complex spectral behavior, which depends on the particle size distribution and optical constants. Photochemical hazes, in particular, are a cause for concern due to uncertainties surrounding their formation and composition and their preference for the cooler, higher metallicity atmospheres of sub-Neptunes - a major class of targets for current and future JWST programs. We propose to build an open source, fast, flexible, and physical model of photochemical hazes that can be used to interpret JWST data for nearly all exoplanets. It will be derived from an oft-used and well validated aerosol microphysics code and updated to take into account new coding practices and advancements in our knowledge of haze physics. The code will be stored on github and open to use by anyone in the exoplanet community for both interpreting data and studying the impact of hazes on atmospheres. We will further use the code to generate a grid of models for a variety of haze and planetary parameters to study trends in haze properties and to facilitate comparisons to data.

OBSERVING DESCRIPTION

Not applicable, as this is an AR Theory proposal.